

# 7. Nonlinear and Adaptive Control of Robotic Manipulators

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The dynamic equations of a robot manipulator in closed form are always written in the form of Euler-Lagrange equations. A dynamic system with  $p$  degrees of freedom can be described by the EL equation as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau$$

where  $q \in \mathbb{R}^p$  is the vector of generalized coordinates representing the joint positions,

$M(q) \in \mathbb{R}^{p \times p}$  is the symmetric positive-definite inertia matrix,  $C(q, \dot{q})\dot{q} \in \mathbb{R}^p$  is

the vector of Coriolis and centrifugal torques,  $g(q) \in \mathbb{R}^p$  is vector of the

gravitational torque, and  $\tau \in \mathbb{R}^p$  is vector of the control torque. And it has the

following properties:

$$q(t) = \begin{bmatrix} q_1(t) \\ \vdots \\ q_p(t) \end{bmatrix}$$

## Properties:

1)  $M(q)$  is positive definite and there exist positive constants  $k_m$  and  $k_m^{-1}$  such that  $k_m x^T x \leq x^T M(q) x \leq k_m^{-1} x^T x$  and  $\|C(x, y)z\| \leq k_c \|x\| \|y\| \|z\|$  for  $x, y, z \in \mathbb{R}^p$ .  
 (反对称)  $A = -A^T \Rightarrow x^T A x = 0, \forall x$ .  
 $\lambda(M - 2C)x = 0$

2)  $\dot{M}(q) - 2C(q, \dot{q})$  is skew symmetric.

→ adaptive control

3) For  $x, y, z \in \mathbb{R}^p$ ,  $M(q)x + C(q, \dot{q})y + g(q) = Y(q, \dot{q}, x, y)\Theta$ , with  $Y(q, \dot{q}, x, y)$

being the regressor and  $\Theta$  being some constant vector.

## 1. Position Control

Control Objective:  $q(t) \rightarrow q_d$ ,  $\dot{q}_d = 0$  (constant)

$$q(t) \rightarrow q_d$$

Define the errors:  $\tilde{q}(t) = q(t) - q_d$ ,  $\dot{\tilde{q}}(t) = \dot{q}(t) - \dot{q}_d = \dot{q}(t)$

The error dynamics:  $M(q)\ddot{\tilde{q}}(t) + C(q, \dot{q})\dot{\tilde{q}}(t) + g(q) = \tau$

$$\Rightarrow M(\tilde{q}(t) + q_d)\ddot{\tilde{q}}(t) + C(\tilde{q} + q_d, \dot{\tilde{q}})\dot{\tilde{q}}(t) + g(\tilde{q}(t) + q_d) = \tau \quad (7.2)$$

Define  $x_1 = \tilde{q}$ ,  $x_2 = \dot{\tilde{q}}$

$$\dot{x} = f(t, x)$$

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \rightarrow 0.$$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = M^{-1}(x_1 + q_d) \left[ \tau - C(x_1 + q_d, x_2) x_2 - g(x_1 + q_d) \right] \end{cases} \rightarrow \text{Autonomous.}$$

Design the following control input using the feedback linearization.

$$\tau = C \cdot x_2 + g - M(k_p \tilde{q} + k_d \dot{\tilde{q}})$$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \tau_0 = -k_p x_1 - k_d x_2 \end{cases}$$

$k_p, k_d > 0.$

With the help of the properties of E-L dynamics, we can design

$$\tau = g(q) - k_p \tilde{q} - k_d \dot{\tilde{q}} \quad (7.3)$$

where  $k_p$  and  $k_d$  are positive definite matrices. The closed-loop system of (7.1) using (7.3) is

$$M(q) \ddot{\tilde{q}} + C(q, \dot{q}) \dot{\tilde{q}} = -k_p \tilde{q} - k_d \dot{\tilde{q}} \quad (7.4)$$

Consider the following Lyapunov function candidate

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$V = \frac{1}{2} \dot{\tilde{q}}^T M(q) \dot{\tilde{q}} = \frac{1}{2} x_2^T M x_2 + \frac{1}{2} \tilde{q}^T k_p \tilde{q}$$

Its derivative is

$$\dot{V} = \dot{\tilde{q}}^T M(q) \ddot{\tilde{q}} + \frac{1}{2} \dot{\tilde{q}}^T \dot{M}(q) \dot{\tilde{q}}$$

$$\stackrel{(7.4)}{=} -\dot{\tilde{q}}^T C \cdot \tilde{q} - \dot{\tilde{q}}^T k_p \tilde{q} - \dot{\tilde{q}}^T k_d \dot{\tilde{q}} + \frac{1}{2} \dot{\tilde{q}}^T \dot{M} \dot{\tilde{q}}$$

$$= \dot{\tilde{q}}^T \left( \frac{1}{2} \dot{M} - C \right) \dot{\tilde{q}} - \dot{\tilde{q}}^T k_p \tilde{q} - \dot{\tilde{q}}^T k_d \dot{\tilde{q}} \quad \checkmark$$

$\stackrel{0}{=} \text{skew symmetric}$

$$= -\dot{\tilde{q}}^T k_d \dot{\tilde{q}} - \dot{\tilde{q}}^T k_p \tilde{q} \quad \checkmark$$

Till now, we can consider the following Lyapunov function candidate

$$V = \frac{1}{2} \dot{\tilde{q}}^T M(q) \dot{\tilde{q}} + \frac{1}{2} \tilde{q}^T k_p \tilde{q}.$$

Its derivative is

$$\dot{V} = -\dot{\tilde{q}}^T K_d \dot{\tilde{q}} \leq 0. \quad \text{NSD.}$$

Note that the closed-loop system is autonomous. We can use the LaSalle's Theorem.

$$E \triangleq \{x \mid \dot{V}=0\} = \{(\tilde{q}, \dot{\tilde{q}}) \mid \dot{\tilde{q}}=0\} = \{x \mid x_2=0\}.$$

Let  $x(t)$  be a solution that belongs identically to  $E$ .

$$x_2=0 \Rightarrow \dot{\tilde{q}}=0 \Rightarrow x_1=0 \\ (\ddot{\tilde{q}}=0) \quad (\tilde{q}=0) \quad \text{from (7.4).}$$

Therefore, the only solution that can stay identically in  $E$  is the origin.

Then, from LaSalle's Theorem, the origin is asymptotically stable, i.e.,

$$\lim_{t \rightarrow \infty} \tilde{q}(t) = 0 \quad \text{and} \quad \lim_{t \rightarrow \infty} \dot{\tilde{q}}(t) = 0.$$

Q2  $M(q)\ddot{q} + (C(q, \dot{q})\dot{q} + D\dot{q} + g(q) = \tau$   
 $\bar{D}$  is psd.

Potential  $P(q)$   
 if  $g(q) = \frac{\partial P(q)}{\partial q}$ .

$$V = \dots + P(q) \\ \dot{P}(q) = \left( \frac{\partial P(q)}{\partial q} \right)^T \cdot \dot{q} \\ = g(q)^T \cdot \dot{q} = \dot{q}^T g(q)$$

## 2. Tracking Control

Control Objective:  $q(t) \rightarrow q_d(t)$ ,  $\dot{q}(t) \rightarrow \dot{q}_d(t)$ ,  $q_d(t)$ ,  $\dot{q}_d(t)$ ,  $\ddot{q}_d(t)$  are bounded.

Define the tracking errors:  $\tilde{q}(t) = q(t) - q_d(t)$ ,  $\dot{\tilde{q}}(t) = \dot{q}(t) - \dot{q}_d(t)$

The error dynamics:  $M(q)(\ddot{\tilde{q}}(t) + \ddot{q}_d(t)) + (C(q, \dot{q})(\dot{\tilde{q}}(t) + \dot{q}_d(t)) + g(q) = \tau$ .

$$\Rightarrow M(q)\ddot{\tilde{q}}(t) + (C(q, \dot{q})\dot{\tilde{q}}(t) = \tau - M(q)\ddot{q}_d - C(q, \dot{q})\dot{q}_d(t) - g(q)$$

Design the following control input:

$$\tau = M(q)\ddot{q}_d + (C(q, \dot{q})\dot{q}_d + g(q)) - K_p \tilde{q} - K_d \dot{\tilde{q}} \quad (7.5)$$

Then the closed-loop system is

$$M(q)\ddot{\tilde{q}} + (C(q, \dot{q})\dot{\tilde{q}}) = -K_p \tilde{q} - K_d \dot{\tilde{q}} \quad (7.6)$$

$\tilde{q}(t) + q_d(t)$        $\dot{\tilde{q}}(t) + \dot{q}_d(t)$        $\rightarrow$  nonautonomous.

Till now, we can consider the following Lyapunov function candidate

$$V = \frac{1}{2} \dot{\tilde{q}}^T M(q) \dot{\tilde{q}} + \frac{1}{2} \tilde{q}^T K_p \tilde{q}. \quad (7.7)$$

Its derivative is

$$\dot{V} = -\dot{\tilde{q}}^T K_d \dot{\tilde{q}} \leq 0. \quad \text{NSD.} \quad (7.8)$$

Since (7.6) is non-autonomous due to the existence of  $q_d(t)$  and  $\dot{q}_d(t)$ , LaSalle's Theorem cannot be used. From (7.8),  $V(t) \leq V(0)$ , which implies that  $\dot{\tilde{q}} \in L_2$  and  $\tilde{q} \in L_\infty$  and  $\tilde{q} \in L_2$ .

Since  $\|(C(q, \dot{q})\dot{\tilde{q}})\| \leq K_c \|\dot{\tilde{q}}\| \cdot \|\dot{\tilde{q}}\| = K_c \|\dot{\tilde{q}} + \dot{q}_d\| \cdot \|\dot{\tilde{q}}\|$ , and  $\dot{q}_d$  is bounded, we can get from (7.6) that  $\ddot{\tilde{q}} \in L_\infty$ . We have  $\dot{\tilde{q}} \in L_2 \cap L_\infty$ ,  $\ddot{\tilde{q}} \in L_\infty$ . From Barbalat's Lemma,  $\lim_{t \rightarrow \infty} \dot{\tilde{q}}(t) = 0$ .

Unfortunately, from the study sketched above, it is not possible to derive any immediate conclusion about the asymptotic behavior of the position error  $\tilde{q}$ .

## 3. Sliding Model Control (for the tracking control)

Define a sliding surface

$$s = \dot{\tilde{q}} + \lambda \tilde{q}, \quad \lambda > 0.$$

$$= \dot{\tilde{q}} - \dot{q}_d + \lambda \tilde{q}$$

$$= \dot{\tilde{q}} - (\underbrace{\dot{q}_d}_{\dot{q}_r} - \lambda \tilde{q}) \triangleq \dot{\tilde{q}} - \dot{q}_r \quad (7.9)$$

$$\Rightarrow \dot{\tilde{q}} = -\lambda \tilde{q} + s \quad (\text{ISS})$$

$\downarrow$  state       $\uparrow$  input

with

$$\dot{\tilde{q}}_r = \dot{q}_d - \lambda \tilde{q} = \dot{q}_d(t) - \lambda (q(t) - q_d(t)) \quad (7.10)$$

Then we have

$$M(q)(\dot{s} + \ddot{q}_r) + C(q, \dot{q})(s + \dot{q}_r) + g(q) = \tau$$

$$\Rightarrow M(q)\dot{s} + C(q, \dot{q})s = \tau - g(q) - M(q)\ddot{q}_r - C(q, \dot{q})\dot{q}_r \quad (7.11)$$

↓  
first-order

Design the following control input

$$\tau = g(q) + M(q)\ddot{q}_r + C(q, \dot{q})\dot{q}_r - Ks \quad (7.12)$$

$\Rightarrow \frac{\gamma(q, \dot{q}, \ddot{q}_r, \dot{q}_r)}{-K \operatorname{sgn}(s)}$

where  $K$  is positive definite. Then the closed-loop system of (7.11) using (7.12) is

$$M(q)\dot{s} + C(q, \dot{q})s = -Ks \quad (7.13)$$

Consider the following Lyapunov function candidate

$$V = \frac{1}{2} s^T M(q) s \quad (7.14)$$

Its derivative along (7.13) is

$$\begin{aligned} \dot{V} &= s^T \dot{M} \dot{s} + \frac{1}{2} s^T \dot{M} s \\ &= -s^T K s - s^T C s + \frac{1}{2} s^T \dot{M} s \\ &= -s^T K s + \frac{1}{2} s^T (\dot{M} - 2C) s \quad \text{skew symmetric.} \\ &= -s^T K s \quad (7.15) \end{aligned}$$

Therefore, the origin  $s=0$  is exponentially stable,  $\lim_{t \rightarrow \infty} S(t) = 0$ .

$$\text{From } s = \dot{\tilde{q}}(t) + \lambda \tilde{q}(t), \Rightarrow \dot{\tilde{q}}(t) = -\lambda \tilde{q}(t) + s \quad (7.16)$$

It follows from  $\lim_{t \rightarrow \infty} S(t) = 0$  that  $\lim_{t \rightarrow \infty} \tilde{q}(t) = 0$  and  $\lim_{t \rightarrow \infty} \dot{\tilde{q}}(t) = 0$ .

$\downarrow$  state,  $\uparrow$  input

Q:  $m(q)\ddot{q} + c(q, \dot{q})\dot{q} + g(q) = \tau + d(t)$   $\|d(t)\| \leq d_{\max} (1 + \|q\|^2 + \|\dot{q}\|)$

$\tau = g(q) + m(q)\ddot{q}_r + c(q, \dot{q})\dot{q}_r - \underline{k} \operatorname{sgn}(s)$   $\xrightarrow{\text{unknown}} \tau$

$\dot{V} \leq -\underline{k}\|s\|_1 + d_{\max}\|s\|_2 \rightarrow s^T \operatorname{sgn}(s) = \|s\|_1$

$\leq -\underline{k}\|s\|_2 + d_{\max}\|s\|_2 \quad (\underline{k} > d_{\max})$

## 7.4 Adaptive Control for Robotic Manipulators

When there exist parametric uncertainties, (7.12) cannot be implemented.

Note that

$$M(q)\ddot{q}_r + c(q, \dot{q})\dot{q}_r + g(q) = Y(q, \dot{q}, \ddot{q}_r, \dot{q}_r) \Theta$$

Since  $\Theta$  is unknown, we can propose the following control input

$$\tau = Y(q, \dot{q}, \ddot{q}_r, \dot{q}_r) \hat{\Theta} - ks \quad (7.14)$$

$$M(q)\dot{s} + c(q, \dot{q})s = \tau - \underbrace{(g(q) + M(q)\ddot{q}_r + c(q, \dot{q})\dot{q}_r)}_{Y(q, \dot{q}, \ddot{q}_r, \dot{q}_r) \Theta} \quad (7.11)$$

Then, using (7.14) for (7.11),

$$\begin{aligned} M(q)\dot{s} + c(q, \dot{q})s &= Y(\hat{\Theta} - \Theta) - ks \\ &= Y\tilde{\Theta} - ks \end{aligned} \quad (7.15)$$

where  $\tilde{\Theta} = \hat{\Theta} - \Theta$ .

Consider the following Lyapunov function candidate

$$V = \frac{1}{2} s^T M s + \frac{1}{2} \tilde{\Theta}^T \Gamma^{-1} \tilde{\Theta}, \quad \Gamma \text{ is PD.} \quad (7.16)$$

Its derivative along (7.15) is

$$\begin{aligned} \dot{V} &= s^T M \dot{s} + \frac{1}{2} s^T \dot{M} s + \tilde{\Theta}^T \Gamma^{-1} \dot{\tilde{\Theta}} \\ &= -s^T k s + \underbrace{s^T Y \tilde{\Theta}}_{= \tilde{\Theta}^T Y^T s \text{ (scalar)}} + \tilde{\Theta}^T \Gamma^{-1} \dot{\tilde{\Theta}} \end{aligned}$$

$$= -s^T K s + \hat{\theta}^T \Gamma^{-1} (\dot{\hat{\theta}} + \tau \Upsilon^T s)$$

Design  $\dot{\hat{\theta}} = -\tau \Upsilon^T s$

(7.17)

We have  $\dot{V} = -s^T K s \leq 0$  (NSD). From Barbalat's Lemma, we can get

$$\lim_{t \rightarrow \infty} s(t) = 0.$$

★ Q:  $M(q)\ddot{q} + C(q, \dot{q}, \dot{q}) + g(q) + d(t) = \tau$ .  
 = design a robust adaptive controller.

## 7.5 Backstepping Control for Robotic Manipulators

Consider the following system for a robotic manipulator

$$\begin{aligned} M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) + \underline{K(q - \theta)} &= 0 & = \tau_1 \\ J\ddot{\theta} + \underline{K(\theta - q)} &= u & = \tau_2 \end{aligned}$$

where  $\underline{q} \in \mathbb{R}^p$  and  $\underline{\theta} \in \mathbb{R}^p$  represent respectively, the vector of link positions and motor angles,  $K$  is the positive diagonal matrix representing the joint stiffness, and

$J$  is the positive diagonal matrix representing the actuator inertia.

Define  $x_1 = \theta$ ,  $x_2 = \dot{\theta}$ . The above dynamics can be described as

$$\boxed{M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) + Kq = Kx_1} \quad (7.18)$$

$$\leftarrow \begin{aligned} M(q)\dot{s} + C(q, \dot{q})s &= -M(q)\ddot{q}_r - C(q, \dot{q})\dot{q}_r - g(q) - Kq + Kx_1 & \dot{x}_1 &= x_2 & (7.19) \end{aligned}$$

$$\dot{x}_2 = J^{-1}u - J^{-1}K(x_1 - q) \quad (7.20)$$

Control Objective:  $q(t) \rightarrow q_d(t)$ ,  $\dot{q}(t) \rightarrow \dot{q}_d(t)$ ,  $q_d(t)$ ,  $\dot{q}_d(t)$ ,  $\ddot{q}_d(t)$  are bounded.

Define:

$$\begin{aligned} \tilde{q} &= q - q_d, & \dot{\tilde{q}} &= \dot{q} - \dot{q}_d \\ \dot{q}_r &= \dot{q}_d - \lambda \tilde{q}, & s &= \dot{\tilde{q}} + \lambda \tilde{q} = \dot{q} - \dot{q}_r \end{aligned}$$

Step 1: Design a feedback control input for  $x_1$  to stabilize the origin  $s=0$ .  
 $\phi_1 = K^{-1} (M(q)\ddot{q}_r + C(q, \dot{q})\dot{q}_r + g(q) + Kq - Ks)$ ,  $K_1$  is PD.

Define  $y_1 = x_1 - \phi_1$ . Then we have

$$M(q)\dot{s} + C(q,\dot{q})s = -K_1 s + K y_1$$

(7.21)

Consider the Lyapunov function candidate

$$V_1 = \frac{1}{2} s^T M s$$

Its derivative is

$$\dot{V}_1 = -s^T K_1 s + s^T K y_1$$

Step 2: Note that the dynamics of  $y_1$  is

$$\dot{y}_1 = \dot{x}_1 - \dot{\phi}_1 = x_2 - \dot{\phi}_1$$

$\phi_2 + y_2$

Consider the following Lyapunov function candidate

$$V_2 = V_1 + \frac{1}{2} y_1^T K y_1$$

(7.22)

Its derivative

$$\dot{V}_2 = \dot{V}_1 + y_1^T K \dot{y}_1$$

$$= -s^T K_1 s + \underline{s^T K y_1} + y_1^T K \dot{y}_1$$

$$= -s^T K_1 s + y_1^T K (\dot{y}_1 + s)$$

$$= -s^T K_1 s + y_1^T K (\underbrace{x_2 - \dot{\phi}_1}_{y_2 + \phi_2} + s)$$

Design a feedback control input for  $x_2$  to stabilize  $y_1 = 0$ ,

$$\phi_2 = \dot{\phi}_1 - y_1 - s, \quad K_2 \text{ is PD.}$$

Define  $y_2 = x_2 - \phi_2$ . Then we have

$$\dot{V}_2 = -s^T K_1 s + y_1^T K (y_2 - y_1)$$

$$= -s^T K_1 s - y_1^T K y_1 + y_1^T K y_2$$

Step 3: The dynamics of  $y_2$  is

$$\begin{aligned}\dot{y}_2 &= \dot{x}_2 - \dot{\phi}_2 \\ &= J^{-1}u - J^{-1}K(x_1 - q) - \dot{\phi}_2\end{aligned}$$

Consider the following Lyapunov function candidate

$$V_3 = V_2 + \frac{1}{2} y_2^T K y_2$$

Its derivative

$$\begin{aligned}\dot{V}_3 &= \dot{V}_2 + y_2^T K \dot{y}_2 \\ &= -s^T K_1 s - y_1^T K y_1 + y_1^T K y_2 + y_2^T K (J^{-1}u - J^{-1}K(x_1 - q) - \dot{\phi}_2)\end{aligned}$$

Design the control input

$$u = J \left( J^{-1}K(x_1 - q) + \dot{\phi}_2 - y_1 - y_2 \right)$$

Then we have

$$\dot{V}_3 = -s^T K_1 s - y_1^T K y_1 - y_2^T K y_2 \leq 0. \quad (NO).$$

From Lyapunov stability theorem,  $\lim_{t \rightarrow \infty} s(t) = 0$ ,  $\lim_{t \rightarrow \infty} y_1 = 0$ ,  $\lim_{t \rightarrow \infty} y_2 = 0$ .

① Introduction of Nonlinear and Adaptive Control.

② Stability of Autonomous system.  $\dot{x} = f(x)$

$V(x)$  p.d.

$\dot{V}$  M.D.  $\rightarrow$  stable  $\rightarrow$  LaSalle's Theorem.

$\downarrow$   
radially unbounded (globally).

$\dot{V}$  M.D. - asy. stable

local stability. linearization

$$J = \frac{\partial f}{\partial x} \Big|_{x=0}$$

③ Stability of Non-autonomous system.  $\dot{x} = f(t, x)$

$$w_1(x) \leq V(t, x) \leq w_2(x)$$

p.d.

decreasing (uniformly)

$\dot{V}$  M.D.  $\rightarrow$  stable

$\dot{V}$  M.D. - asy. stable

$\rightarrow$  Barbălat's Lemma

LaSalle - Yoshizawa Theorem

BLBO

ISS.

④ Adaptive Control

Direct and Indirect MRAC design for scalar systems

Direct MRAC for MIMO systems.

Robust Adaptive Control: dead zero,  $\sigma$ -modification,  $\epsilon$ -modification, (adaptive)

⑤ Nonlinear and Adaptive Control design.

Feedback linearization

sliding mode control

Backstepping

⑥ Control of Robotic Manipulators.